

Assignment#4: Numerical Solution of Flow through CD Nozzle using van Leer Flux Vector Splitting Method

Problem Statement

Solve the quasi-1-D Euler equations for the isentropic flow of a calorically perfect gas with $\gamma = 1.4$ through a nozzle (as shown in Figure 1) having an area distribution

$$A = 1.0 + 2.0(x - 1)^2, \quad 0 \leq x \leq 2$$

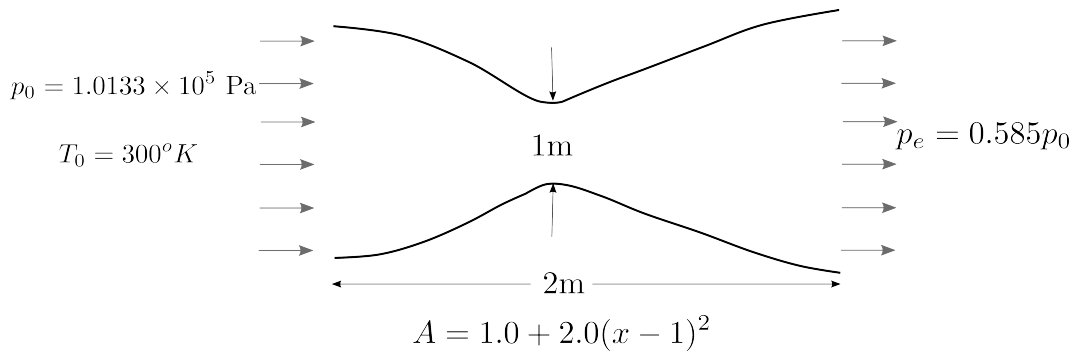


Figure 1: Flow through CD Nozzle

Use van Leer Flux Vector Splitting method to solve the governing equations for solve for an exit to stagnation pressure ratio, p_e/p_0 , of 0.585. Use a reservoir temperature and pressure as 300 K and $1.0133 \times 10^5 \text{ Pa}$ respectively.

Plot (a) the nondimensional pressure, p/p_0 , versus x ; (b) Mach number versus x . (c) Use your gasdynamics knowledge to compute the exact quasi-1D results and compare with your numerical results.

Governing Equations

For quasi-one-dimensional (quasi-1D) nozzle flows, the governing Euler equations describe the conservation of mass, momentum, and energy while accounting for variations in cross-sectional area. The equations are derived from the full three-dimensional Euler equations under the assumption of axial symmetry and small flow angle variations, leading to the following conservative form:

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}}{\partial x} = \mathbf{S} \quad (1)$$

where the conservative (state vector), flux vector and the source terms are given as

$$\mathbf{U} = \begin{bmatrix} \rho A \\ \rho u A \\ EA \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} \rho u A \\ (\rho u^2 + p)A \\ u(E + p)A \end{bmatrix}, \quad \mathbf{S} = \begin{bmatrix} 0 \\ p \frac{\partial A}{\partial x} \\ 0 \end{bmatrix}$$

Here, $\frac{dA}{dx}$ represents the area variation along the nozzle.

Numerical Scheme

The van-Leer Flux Vector Splitting (FVS) methods[1] to be used for solving the quasi-1D Euler equations (1) to get the solution of the CD nozzle problem is described above.

Discretized Governing Equation

Applying first-order upwind scheme to the Governing equations (1), we get

$$\mathbf{U}_i^{n+1} = \mathbf{U}_i^n - \frac{\Delta t}{\Delta x} (\mathbf{F}_i^{+n} - \mathbf{F}_{i-1}^{+n}) - \frac{\Delta t}{\Delta x} (\mathbf{F}_{i+1}^{-n} - \mathbf{F}_i^{-n}) + \Delta t \mathbf{S}_i \quad (3)$$

Where $\mathbf{F}^+$ and $\mathbf{F}^-$ can be determined using van Leer FVS splitting strategy[1], for which split fluxes are given below. Note, you need to consider all the possible ranges of Mach number in your computer code as specified below. The discretized form of the source term can be considered as

$$\mathbf{S}_i = \begin{bmatrix} 0 \\ p_i \frac{A_{i+1} - A_{i-1}}{2\Delta x} \\ 0 \end{bmatrix}$$

Split Fluxes for van Leer Method

1. For $M \leq -1$,

$$\mathbf{F}_{VL}^+ = 0 \quad \text{and} \quad \mathbf{F}_{VL}^- = \mathbf{F}$$

2. For $-1 < M < 1$,

$$\mathbf{F}_{VL}^+ = \frac{1}{4}\rho a(M+1)^2 A \begin{bmatrix} 1 \\ \frac{2a}{\gamma} \left(1 + \frac{\gamma-1}{2}M\right) \\ \frac{2a^2}{\gamma^2-1} \left(1 + \frac{\gamma-1}{2}M\right)^2 \end{bmatrix} \quad \mathbf{F}_{VL}^- = \mathbf{F}_{VL} - \mathbf{F}_{VL}^+$$

3. For $M \geq 1$,

$$\mathbf{F}_{VL}^+ = \mathbf{F} \quad \text{and} \quad \mathbf{F}_{VL}^- = 0$$

where $a = \sqrt{\gamma RT}$ is the speed of sound in the medium. The ratio of specific heats for air $\gamma = 1.4$.

Time Step from Stability Condition

The time step Δt for time-marching is calculated as

$$\Delta t = \nu \frac{\Delta x}{\lambda_{max}}$$

where

$$\lambda_{max} = \max \{(|u[1]| + a[1]), (|u[2]| + a[2]), \dots, (|u[N]| + a[N])\}$$

with $I_{max}=101$ and ν is CFL number. For stability $\nu \leq 1$.

Grids & Initialization

Use $I_{max}=101$ grid points along the nozzle. Initialize the whole flow field based on inlet condition.

Boundary Treatment

At Inlet: The stagnation temperature and stagnation pressure at the inlet boundary are assumed to be equal to the reservoir temperature and pressure. The inlet velocity is extrapolated from the interior grid point (by setting $u_1 = u_2$, where u_1 is the velocity at the inlet boundary and u_2 is the velocity at the nearest interior grid point). Specify consistent temperature, pressure and density from isentropic relation based on the inlet stagnation temperature and stagnation pressure.

At Outlet: The exit pressure p_e is specified using the exit-to-stagnation pressure ratio, $p_e/p_0 = 0.585$. Density and velocity are extrapolated from the interior domain, following the conditions $\rho_{I_{max}} = \rho_{I_{max}-1}$ and $u_{I_{max}} = u_{I_{max}-1}$.

References

- [1] Charles Hirsch, *Numerical Computation of Internal and External Flows*, (Section 20.2.3 pages 420-425) Volume II, Wiley, 1990.

- **Checklist for submission:**

1. Submit only the original source code written in C, C++, or Python, with proper inline documentation for each function. Use meaningful variable names to enhance readability.

- **Important Note:**

- Submitting code in any other programming language or as a .pdf, .txt, or .docx file will result in zero marks for the entire assignment, as these formats cannot be compiled.
 - Online compiler should not be used for your code.
 - 2. Include a README.txt file with clear instructions on how to run your code and generate the required plots.
 - 3. Report: Submit a brief report in PDF format, including:
 - Provide a flowchart representing the logic of your program.
 - All relevant results presented as plots with explanations.
 - A comparison between the van Leer FVS results and the exact solution.
 - The plots you submit must be independently reproducible from your code by the teaching assistants (TAs).

- **Instruction for submission:**

- Rename your program file using your roll number (e.g., 204010006.c).
 - Rename your report file using your roll number (e.g., 204010006.pdf).
 - Upload all three required documents separately, as specified in the checklist above.
 - Don't submit a zip file.

- **Important Notes:**

- Refer to the Rubric on Moodle before submission.
 - Marks will be awarded only if the program runs correctly and produces the expected results as per the rubric.
 - Submissions not following the instructions will not be evaluated.

- **Strict Warning: Plagiarism is strictly prohibited!** Copying code from others, online sources, or previous submissions will result in **severe penalties**.

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